

Design Document: SRAM256B

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Key Features

Table – 1: SRAM256B Features

Word Capacity	256
Number of Bits per word	8
Address Width	8
Core Area (um ²)	2128.159
Flip-Flop Area	85.4%
Logic Area	14.6%
Gate Count	3881
Core Delay (ps)	101
Pin-2-Pin Delay (ps)	NA
Leakage Power (uw)	11.856
Dynamic Power (uw)	115.283
Technology	TSMC12NM

Functional Description

The Static RAM has 256 words and 8-bit word length. Each 8-bit word can be accessed using one 8-bit address. This block includes the following functional features:

- Chip enable control enable controls the operations of the block.
- The reset control drives the output bus to 00h and operates in synchronous with the rising edge of the system clock.
- The output holds the last read out value during the write and no operation state.
- The ready signal differentiates between busy mode and new access.

The features can be extended to

- Transparent write
- Bulk erase

SRAM Block Diagram

The SRAM has control signals. The block diagram is shown in Figure – 1.

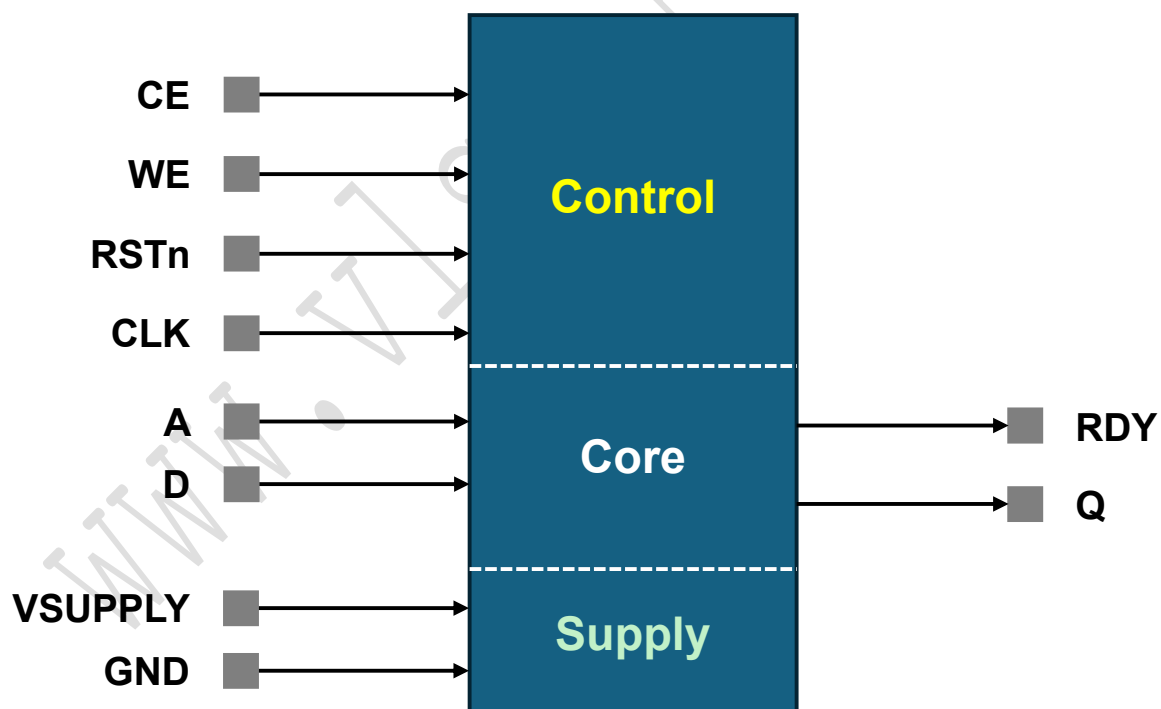


Figure – 1: SRAM Block Diagram

Pin Description

Table – 1: SRAM Pin Description

Pin	Direction	Description
CE	Input	Chip Enable: CE HIGH (positive edge) activates read or write access.
WE	Input	Write Enable: WE (along with CE) define the command being entered. (WE = LOW → Read Access, WE = HIGH → Write Access)
RSTn	Input	Reset: An active-LOW signal. RSTn must be HIGH during the normal operation. LOW state will reset the output bus without performing the read or write access. This is synch
CLK	Input	Clock Signal: The SRAM operates at the rising edge of the CLK signal. This is derived from the system clock to synchronize the operations.
A [7:0]	Input	Address Inputs: Address for read and write commands to select one location out of the memory array.
D [7:0]	Input	Data Inputs: The input data for write command. During the read operation, data is ignored.
Q [7:0]	Output	Data Outputs: Data output for read command. Transparent write operation can be implemented as an extended feature. Data outputs are latched.
RDY	Output	Ready Signal: RDY HIGH indicates the ready state of SRAM to start any new access. RDY LOW indicates busy state of SRAM.

Operation Modes

Table – 2: Control Signals and Modes

CE	RSTn	WE	D	Operation
L	X	X	X	No Operation (NOP)
H	L	X	X	Reset
H	H	H	Data	Write
H	H	L	X	Read

Operations and Timing Diagram

Reset Mode

The output bus goes to reset state when chip enable goes to HIGH and reset control goes LOW. This is shown in Figure – 2 during T3 to T5. After reset the write or read operation can be performed depending on, WE control and the address in address bus.

A read access after the reset mode will result a valid data from the selected locations in consecutive cycles.

A write access after the reset mode will hold 00h.

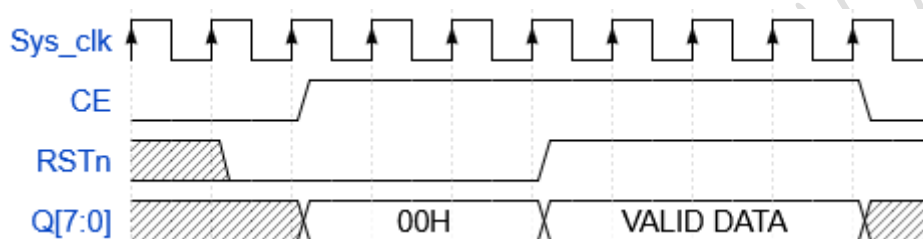


Figure – 2: Reset Mode Operation

Write Mode

The write access is performed by driving WE = HIGH. The chip enable must be HIGH and RSTn = HIGH.

The first write operation will take place at T4 writing FFh to the address 46h and the consecutive cycles. The write operations to the consecutive locations can be verified with read access followed by the write access.

The output bus is hold to 00H as the write access starts immediate after reset mode. The output bus will hold the last read out value when write access is performed after read access.

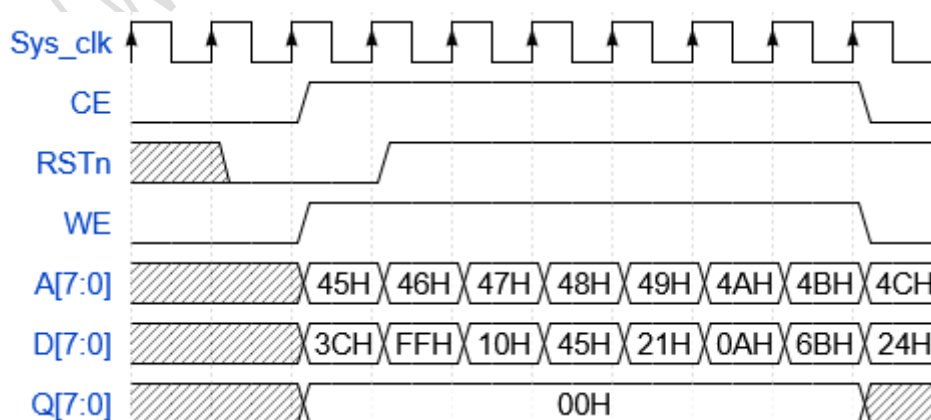


Figure – 3: Write Mode Operation

Read Mode

The read operation is performed after a write operation so that the valid data will be placed in the output bus otherwise garbage data will be read out.

When WE = LOW at T10, the content of 4Ch i.e. 45h is placed in the output data bus. The consecutive locations are read out till the WE signal is LOW representing the read mode.

The data in the input data bus will not affect the memory location.

When RSTn is asserted, output bus resets to 00h as long as CE is HIGH.

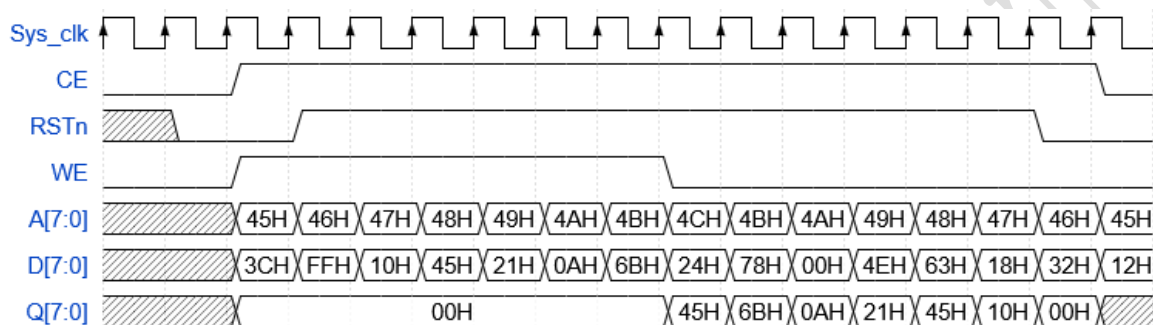


Figure – 4: Multi-Cycle Read Operation